12 November 2026

Dear Editor,

Please consider our manuscript, *Separating coverage gains from error targeting in molecular prediction sets*, as an original research article in *Chemometrics and Intelligent Laboratory Systems*. This is a substantially revised new submission following the decision on CHEMOLAB-D-26-01118.

The study addresses a practical problem in evaluating molecular uncertainty: expanding a prediction set can improve coverage simply by withholding more decisions. We introduce a comparison that fixes the predictor and the number of expanded decisions within each original singleton class, together with an exact condition for when weighted expansion adds a label.

The completed evaluation contains 960 predictor units across twelve Tox21 assays and fixed random/scaffold holdouts from four additional assay families. It reproduces the historical minimum-class coverage increase of 1.3649 percentage points and shows that a simple omitted-label-probability rule reaches the same gain at the same deferral budget. The activation analysis further distinguishes passing transport diagnostics from changing a decision and recovering an error. These findings turn an aggregate improvement into a concrete account of decision cost and information content.

The revision adds shared-predictor and class-budget comparisons, complete reference-system evaluations, component and gate sensitivities, dependence-aware paired intervals, and executable manuscript reconstruction. The main text and supporting information are accompanied by a point-by-point response explaining the substantive changes from the earlier submission.

We believe the resulting comparison framework and molecular case study will interest readers concerned with reliable chemometric prediction, uncertainty assessment, and the practical value of model complexity.

Thank you for considering our manuscript.

Sincerely,

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On behalf of the authors

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